CO2 – AT1

Artificial Intelligence – MCQ Questions and Answers

Name: Lakshana K
Register Number: 192524433
Course Code: CSA11712
Course Name: Artificial Intelligence

Questions with Answers

1. In informed search strategies, some algorithms use only the heuristic value to decide which node to expand next. This approach focuses on selecting the node that appears closest to the goal without considering the cost already incurred. Which algorithm follows this principle?

A) Breadth First Search
B) Uniform Cost Search
C) Greedy Best-First Search
D) Depth First Search

Answer: C) Greedy Best-First Search

2. A search is widely used in pathfinding and graph traversal because it combines the actual cost from the start node and the estimated cost to the goal. What is the evaluation function used in A search?**

A) f(n) = g(n)
B) f(n) = h(n)
C) f(n) = g(n) + h(n)
D) f(n) = g(n) × h(n)

Answer: C) f(n) = g(n) + h(n)

*3. Heuristic functions play a crucial role in informed search by guiding the algorithm toward the goal. What property ensures optimality in A search?**

A) Completeness
B) Admissibility
C) Depth limit
D) Backtracking

Answer: B) Admissibility

4. Which of the following is a local search algorithm?

A) A* Search
B) Hill Climbing
C) Breadth First Search
D) Uniform Cost Search

Answer: B) Hill Climbing

5. In adversarial search, agents compete against each other in a game environment. Which algorithm is commonly used for such decision-making?

A) Greedy Search
B) Minimax Algorithm
C) Backtracking
D) Uniform Search

Answer: B) Minimax Algorithm

6. A delivery robot selects paths based only on the estimated distance to the destination without considering obstacles already encountered. Which algorithm is the robot using?

A) A* Search
B) Greedy Best-First Search
C) Uniform Cost Search
D) Depth Limited Search

Answer: B) Greedy Best-First Search

7. A GPS navigation system combines the distance already travelled and an estimate of the remaining distance to the destination. Which algorithm best represents this approach?

A) Depth First Search
B) A* Search
C) Hill Climbing
D) Backtracking

*Answer: B) A Search**

8. A university is scheduling exams with constraints such as no overlapping exams and limited rooms. This problem is best modeled as:

A) Game Playing Problem
B) Constraint Satisfaction Problem
C) Heuristic Search Problem
D) Optimization Problem

Answer: B) Constraint Satisfaction Problem

9. An AI-based chess program evaluates possible moves and anticipates the opponent's responses before making a decision. Which algorithm is being applied?

A) Greedy Search
B) Hill Climbing
C) Minimax Algorithm
D) Breadth First Search

Answer: C) Minimax Algorithm

10. A robot explores an unknown environment and makes decisions step-by-step while updating its knowledge. Which type of agent is suitable?

A) Offline Agent
B) Online Search Agent
C) Static Agent
D) Deterministic Agent

Answer: B) Online Search Agent

*11. Why is an admissible heuristic important in A search?**

A) It reduces memory usage
B) It ensures optimality
C) It increases branching factor
D) It avoids loops

Answer: B) It ensures optimality

12. What is the primary benefit of Alpha-Beta pruning?

A) Reduces depth of tree
B) Reduces number of nodes evaluated
C) Changes game rules
D) Improves heuristic accuracy

Answer: B) Reduces number of nodes evaluated

13. What is the main drawback of Hill Climbing?

A) High memory usage
B) Revisiting nodes
C) Getting stuck in local maxima
D) Requires complete tree

Answer: C) Getting stuck in local maxima

14. What ensures a valid CSP solution?

A) Heuristic function
B) Path cost
C) Constraint satisfaction
D) Depth-first traversal

Answer: C) Constraint satisfaction

15. What is the role of an evaluation function in adversarial games?

A) Generate successors
B) Estimate utility of a state

C) Reduce branching factor
D) Store moves

Answer: B) Estimate utility of a state

*16. In A search, suppose a node has a path cost of 7 and a heuristic estimate of 5. What is the value of f(n)?**

A) 12
B) 2
C) 35
D) 7

Answer: A) 12

Calculation:
f(n) = g(n) + h(n)
f(n) = 7 + 5
f(n) = 12

17. A heuristic function overestimates the actual cost to reach the goal. What happens in such a case?

A) A* remains optimal
B) A* becomes non-optimal
C) A* becomes BFS
D) A* stops working

*Answer: B) A becomes non-optimal**

18. Consider a search tree with a branching factor of 2 and depth of 4. What is the approximate number of nodes?

A) 10
B) 15
C) 31
D) 16

Answer: C) 31

Calculation:
Total nodes = 1 + 2 + 4 + 8 + 16
Total nodes = 31

19. In a minimax tree, each node has 3 children and the depth is 3. How many leaf nodes are present?

A) 9
B) 27

C) 6
D) 12

Answer: B) 27

Calculation:
Leaf nodes = b^d
= 3^3
= 27

20. What is the best-case time complexity of Alpha-Beta pruning?

A) O(b^d)
B) O(b^(d/2))
C) O(d^b)
D) O(log b)

Answer: B) O(b^(d/2))